Joint Model Importance via the Rashomon Set

Behavioural Comparison of Near-Equally-Good Models

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Beyond the single best model

- In tabular ML, many models reach indistinguishable predictive performance.
- The *Rashomon set* $\mathcal{R}_\varepsilon = \{\, m : \mathrm{score}(m) \leq \mathrm{score}(m^*) \cdot (1+\varepsilon) \,\}$ collects every model within ε of the best.
- Members of $\mathcal{R}_\varepsilon$ can disagree substantially on individual predictions and on feature importance.
- **Question.** How are these equally good models related to one another, and what does the collective tell us beyond the single best?

Pipeline

1. For every $m \in \mathcal{R}_\varepsilon$, compute predictions on a held-out 2/3 validation split.
2. Build the pairwise *behavioural distance* matrix $d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p$.
3. View the matrix three ways:
    - hierarchical clustering (`hclust`)
    - classical MDS (`cmdscale`, $k = 2$)
    - partitioning around medoids (`cluster::pam`)

Predictions are class probabilities for classification, $\hat{y}$ for regression. Distances computed for both Euclidean and Manhattan; results shown are Euclidean (Manhattan tells the same story).

Setup

Rashomon tolerance $\varepsilon = 1\%$ of the best score.
Five learner families considered for every task: `tree`, `glmnet`, `xgb`, `nnet`, `svm`.

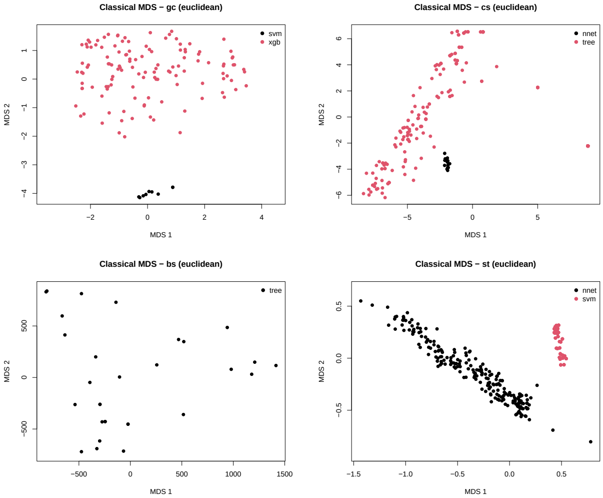
Key	Task	Type
`gc`	German Credit	classification
`cs`	COMPAS	classification
`bs`	Bike Sharing	regression
`st`	Synthetic	regression

Who survives the Rashomon cut?

Task	Surviving families (n models)
`gc`	`svm` (9), `xgb` (125)
`cs`	`nnet` (15), `tree` (261)
`bs`	`tree` (30)
`st`	`nnet` (216), `svm` (160)

First finding. At $\varepsilon = 1\%$ only one or two learner families survive per task — *learner choice* is the first dimension of disagreement inside $\mathcal{R}_\varepsilon$.

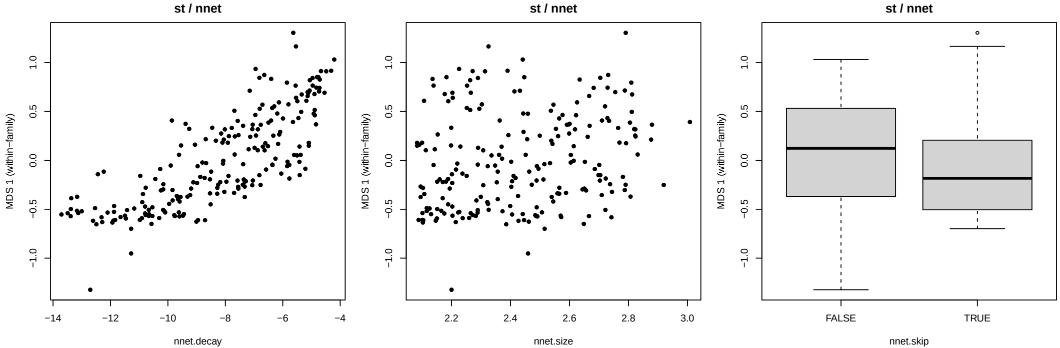
MDS view — all four tasks



Reading the MDS plots

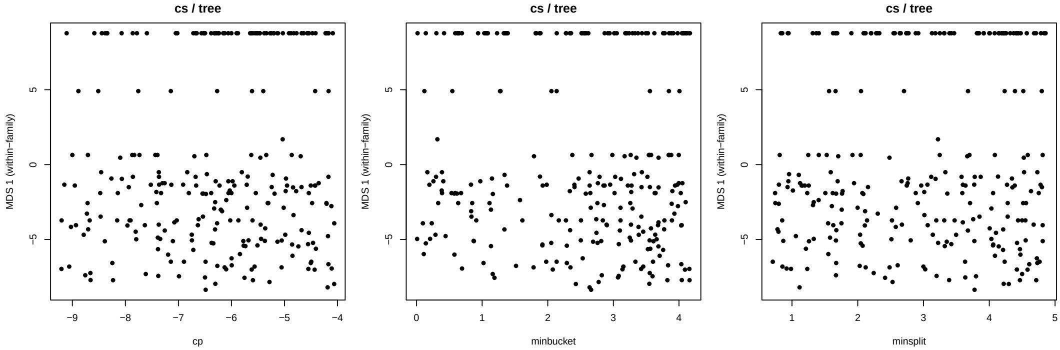
- Wherever two families coexist, behavioural distance **separates them cleanly** — the dominant axis is family identity, not within-family variation.
- Within-family variance is *asymmetric*: `svm` and `nnet` form tight clusters; `tree` and `xgb` scatter widely despite fewer tunable hyperparameters.
- Several families trace out a **1D-manifold** in the MDS plane (e.g. `st/nnet`, `cs/tree`) — next two slides.

Within-family structure I — st / `nnet`



- Each panel: 1D within-family MDS coordinate vs. one hyperparameter.
- The within-family manifold **is the `decay` axis** — near-monotone mapping.
- `size` and `skip` have no effect on behavioural position.

Within-family structure II — cs / tree



- Within-family MDS coordinate vs. `cp`, `minbucket`, `minsplit`.
- Trees on COMPAS are **bimodal**: a tight cluster at MDS 1 ≈ 8 and a diffuse population below zero.
- *No single tuned hyperparameter* separates the two modes — a discrete behavioural split that tuning alone does not reach.

First interpretation

- **Joint model importance is a two-level question:**
  1. Which learner family does the model come from?
  2. Where on the family's behavioural manifold does it sit?
- Across every task observed, learner choice is the first axis of disagreement — not hyperparameter setting.
- Within-family structure can be smooth and HP-aligned (`st/nnet` $\sim$ `decay`) or discrete and HP-orthogonal (`cs/tree`).

Next steps

- ε **sensitivity.** How does family composition shift as the Rashomon tolerance opens?
- **Feature-importance distance.** Compare behavioural distance with a distance built from VIC — do the clusters agree?
- **Choose k for PAM via silhouette**, then read out the medoids as "representative" members of the Rashomon set.
- **Disagreement map.** Where in feature space do Rashomon-set models disagree the most?
- **Discrete tree mode on `cs`.** Is the bimodality driven by structural differences (root split, depth) that hyperparameters do not capture?

Thank you.

Questions?